SAT Math Trick Pack 3

68 new challenge items · Answers and solutions

Name ..... Date .....

Each question states its tool: **No calculator needed**, **Calculator** or **Desmos**. Multiple-choice questions have one correct answer. For student-produced responses (SPR), write your answer in the box.

Algebra (6 items)

1. A-337 · Hard · Multiple choice · **No calculator needed**

The solution to the system $ax + by = 31$ and $bx + ay = 19$, where a and b are constants, is $(4, 1)$. What is the value of $a^2 - b^2$?

A) 16

B) 40

C) 10

D) 58

Answer: (B) 40

$4a + b = 31$ and $a + 4b = 19$. Adding: $5a + 5b = 50$, so $a + b = 10$. Subtracting: $3a - 3b = 12$, so $a - b = 4$. $a^2 - b^2 = (a + b)(a - b) = 40$ ($a = 7$, $b = 3$).

Trap: Students report $a + b = 10$, $(a - b)^2 = 16$ or $a^2 + b^2 = 58$.

---

2. A-338 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the graph of $3x - ky = 12$, where k is a nonzero constant, has an x-intercept equal to its y-intercept. What is the value of k?

A) -4

B) 3

C) 4

D) -3

Answer: (D) -3

x-intercept: $3x = 12$, $x = 4$. y-intercept: $-ky = 12$, $y = -\dfrac{12}{k}$. Set $-\dfrac{12}{k} = 4$: $k = -3$.

Trap: Students drop the negative sign in $-ky$ and get $k = 3$.

---

3. A-339 · Hard · Multiple choice · **No calculator needed**

For every real number m, the graph of $y = mx + 3m - 2$ in the xy-plane passes through the point P. What are the coordinates of P?

A) $(-3, -2)$

B) $(0, -2)$

C) $(3, -2)$

D) $(-2, 3)$

Answer: (A) $(-3, -2)$

Rewrite: $y + 2 = m(x + 3)$. This holds for every m only when $x + 3 = 0$ and $y + 2 = 0$, so $P = (-3, -2)$.

Trap: Students test only $m = 0$ and pick $(0, -2)$, or make a sign error.

---

4. A-340 · Hard · Multiple choice · **No calculator needed**

A cafe sells juice for 4 dollars each and sandwiches for 7 dollars each. A customer spends exactly 112 dollars and buys at least one juice and at least one sandwich. How many different combinations of juices and sandwiches are possible?

A) 5

B) 4

C) 3

D) 28

Answer: (C) 3

$4x + 7y = 112$ with $x, y \geq 1$ integers. $4x = 112 - 7y$ needs y to be a multiple of 4: $y = 4, x = 21$; $y = 8, x = 14$; $y = 12, x = 7$ ($y = 16$ gives $x = 0$, not allowed). 3 combinations.

Trap: Students include $(28, 0)$ or $(0, 16)$ (answers 4 or 5) or divide 112 by 4.

---

5. A-341 · Medium · Multiple choice · **No calculator needed**

Which inequality has the solution set $x \leq -3$ or $x \geq 11$?

A) $|x + 4| \geq 7$

B) $|x - 4| \leq 7$

C) $|x - 4| \geq 7$

D) $|x - 7| \geq 4$

Answer: (C) $|x - 4| \geq 7$

The centre of -3 and 11 is 4 and the distance to each is 7; the solutions lie outside: $|x - 4| \geq 7$.

Trap: Students reverse the inequality, flip the sign of the centre, or swap centre and distance.

---

6. A-342 · Hard · Multiple choice · **No calculator needed**

For how many integer values of k does the point (k, k) satisfy both $y \geq 2x - 5$ and $y > -x + 1$?

A) 5

B) 4

C) Infinitely many

D) 6

Answer: (A) 5

$k \geq 2k - 5 \Rightarrow k \leq 5$. $k > -k + 1 \Rightarrow k > \frac{1}{2}$. Integers $1, 2, 3, 4, 5$: 5 values.

Trap: Students treat $\geq$ as $>$ (getting 4) or use only one inequality.

Advanced Math (10 items)

7. AM-322 · Hard · Multiple choice · **No calculator needed**

In the equation $\dfrac{ax}{x - 2} = 3 + \dfrac{4}{x - 2}$, a is a constant. For which values of a does the equation have no solution?

A) 3 and 4 only

B) 3 only

C) 2 only

D) 2 and 3 only

Answer: (D) 2 **and** 3 **only**

Multiply by $x - 2$: $ax = 3(x - 2) + 4$, so $(a - 3)x = -2$. If $a = 3$: $0 = -2$, no solution. Otherwise $x = \dfrac{-2}{a - 3}$, which is extraneous when it equals 2: $a - 3 = -1$, $a = 2$. So $a = 2$ and $a = 3$.

Trap: Students find only the coefficient case ($a = 3$) and forget the extraneous-root case.

---

8. **AM-323** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 3x^2 - 5x + 2$. Which expression is equivalent to $f(a+1) - f(a)$?

A) 0

B) $6a - 2$

C) $6a + 3$

D) $6a + 8$

Answer: (B) $6a - 2$

$f(a+1) = 3(a+1)^2 - 5(a+1) + 2 = 3a^2 + a$. $f(a) = 3a^2 - 5a + 2$. Difference: $6a - 2$.

Trap: Students assume $f(a+1) = f(a) + f(1)$, giving 0 because $f(1) = 0$, or drop the linear term.

---

9. **AM-324** · Hard · SPR · **No calculator needed**

For how many integer values of b does the equation $x^2 + bx + 36 = 0$ have two distinct integer solutions?

Answer: 8

The roots p, q satisfy $pq = 36$ and $b = -(p+q)$. Distinct factor pairs: $(1, 36), (2, 18), (3, 12), (4, 9)$ and their negatives, giving $b = \pm 37, \pm 20, \pm 15, \pm 13$: 8 values. $(6, 6)$ is a repeated root.

Trap: Students count positive b only (4) or include $b = \pm 12$ (10).

---

10. **AM-325** · Hard · Multiple choice · **No calculator needed**

The equations $x^2 - x - 12 = 0$ and $x^2 + kx + 8 = 0$ have a common solution, where k is a positive constant. What is the value of k?

A) -6

B) $\dfrac{17}{3}$

C) $-\dfrac{17}{3}$

D) 6

Answer: (B) $\dfrac{17}{3}$

$x^2 - x - 12 = (x-4)(x+3)$. Common root 4: $16 + 4k + 8 = 0$, $k = -6$ (not positive). Common root -3: $9 - 3k + 8 = 0$, $k = \frac{17}{3}$.

Trap: Students test only the first root and answer -6, or ignore the condition $k > 0$.

---

11. **AM-326** · Hard · Multiple choice · **Desmos**

What is the sum of all distinct real solutions of $x^3 - 2x^2 - 15x + 36 = 0$?

A) 2

B) 7

C) -2

D) -1

Answer: (D) -1

$x^3 - 2x^2 - 15x + 36 = (x-3)^2(x+4)$. The distinct solutions are 3 and -4; sum $= -1$. (Vieta's sum, 2, counts the double root twice.) Desmos: the graph touches the x-axis at 3 and crosses at -4.

Trap: Students use the sum of roots from the coefficients (2) without checking for a repeated root.

---

12. **AM-327** · Hard · Multiple choice · **No calculator needed**

What is the solution set of the equation $2x^{2/3} - 3 = 15$?

A) $\{-27, 27\}$

B) $\{27\}$

C) $\{-9^{2/3}, 9^{2/3}\}$

D) $\{-6, 6\}$

Answer: (A) $\{-27, 27\}$

$x^{2/3} = 9 \Rightarrow \left(x^{1/3}\right)^2 = 9 \Rightarrow x^{1/3} = \pm 3 \Rightarrow x = \pm 27$. Check: $(-27)^{2/3} = \left((-27)^{1/3}\right)^2 = 9$.

Trap: Students keep only the positive root, or invert the exponent the wrong way.

---

13. **AM-328** · Hard · Multiple choice · **No calculator needed**

The functions f and g are defined by $f(x) = 7 \cdot 4^x$ and $g(x) = f(x - 1.5)$. For all values of x, $g(x) = k \cdot f(x)$, where k is a constant. What is the value of k?

A) -6

B) $\frac{7}{8}$

C) $\frac{1}{8}$

D) 8

Answer: (C) $\frac{1}{8}$

$g(x) = 7 \cdot 4^{x-1.5} = 7 \cdot 4^x \cdot 4^{-1.5} = f(x) \cdot \frac{1}{8}$. So $k = \frac{1}{8}$.

Trap: Students shift in the wrong direction ($k = 8$) or subtract 1.5 from the base.

---

14. **AM-329** · Hard · Multiple choice · **No calculator needed**

The function f is defined by $f(x) = 2(x-1)(x-7)$. For what value of k does the graph of $y = f(x) + k$ have exactly one x-intercept?

A) 9

B) 14

C) 18

D) -18

Answer: (C) 18

The vertex is midway between the zeros, at $x = 4$: $f(4) = 2(3)(-3) = -18$. Shifting up by 18 makes the vertex touch the x-axis: $k = 18$.

Trap: Students answer the minimum value -18, or forget the leading coefficient (9).

---

15. **AM-330** · Hard · Multiple choice · **No calculator needed**

The function g is defined by $g(x) = 3^x$. If $g(a) = 4$, what is the value of $g(2a - 1)$?

A) $\frac{16}{3}$

B) 15

C) $\frac{8}{3}$

D) 7

Answer: (A) $\frac{16}{3}$

$g(2a-1) = 3^{2a-1} = \frac{(3^a)^2}{3} = \frac{16}{3}$.

Trap: Students apply the input operations to the output ($2 \cdot 4 - 1 = 7$) or compute $4^2 - 1$.

---

16. **AM-331** · Medium · Multiple choice · **No calculator needed**

If $2a - 3b = 7$, what is the value of $\dfrac{4^a}{8^b}$?

A) $\dfrac{1}{128}$

B) 2^{14}

C) 7

D) 128

Answer: (D) 128

$\dfrac{4^a}{8^b} = \dfrac{2^{2a}}{2^{3b}} = 2^{2a-3b} = 2^7 = 128.$

Trap: Students answer 7, or subtract the exponents in the wrong order.

Geometry and Trigonometry (17 items)

17. **GT-258** · Hard · Multiple choice · **No calculator needed**

Right triangles PQR and XYZ are similar, with P, Q and R corresponding to X, Y and Z, and right angles at Q and Y. If $\tan P = \dfrac{8}{15}$ and $XZ = 51$, what is the length of $\overline{XY}$?

A) 27.2

B) 45

C) 24

D) 95.625

Answer: (B) 45

$\tan X = \tan P = \dfrac{8}{15} = \dfrac{YZ}{XY}$, so the sides are $8k$, $15k$ and hypotenuse $17k$. $17k = 51$, $k = 3$, $XY = 15k = 45$.

Trap: Students give the opposite side (24) or multiply the hypotenuse by the tangent (27.2) or divide by it (95.625).

---

18. **GT-259** · Hard · Multiple choice · **No calculator needed**

In triangle ABC, $AB = AC = 17$ and $BC = 16$. What is the value of $\tan B$?

A) $\dfrac{8}{15}$

B) $\dfrac{15}{8}$

C) $\dfrac{15}{17}$

D) $\dfrac{16}{17}$

Answer: (B) $\dfrac{15}{8}$

The altitude from A bisects $\overline{BC}$: half-base 8, altitude $\sqrt{17^2 - 8^2} = 15$. $\tan B = \dfrac{15}{8}$.

Trap: Students use the whole base, flip the ratio, or give $\sin B$.

---

19. GT-260 · Hard · Multiple choice · **No calculator needed**

A rectangular box has a base that measures 9 by 12 and a height of 20. Let θ be the angle between a diagonal of the box (from a bottom corner to the opposite top corner) and the base. What is the value of $\tan\theta$?

A) $\frac{20}{9}$

B) $\frac{5}{3}$

C) $\frac{3}{4}$

D) $\frac{4}{3}$

Answer: (D) $\frac{4}{3}$

The diagonal's projection on the base is the base diagonal, $\sqrt{9^2+12^2}=15$. $\tan\theta=\frac{20}{15}=\frac{4}{3}$.

Trap: Students use one base edge (9 or 12) instead of the base diagonal, or flip the ratio.

---

20. GT-261 · Hard · Multiple choice · **No calculator needed**

In right triangle ABC, angle C is the right angle and $\cos A=k$. Which expression is equal to $\sin^2 A+\sin^2 B$?

A) 1

B) The value cannot be determined from the information given.

C) $2-2k^2$

D) $2k^2$

Answer: (A) 1

A and B are complementary, so $\sin B=\cos A$. $\sin^2 A+\sin^2 B=\sin^2 A+\cos^2 A=1$.

Trap: Students think the value depends on k or cannot be found.

---

21. GT-262 · Hard · Multiple choice · **No calculator needed**

Right triangle ABC has its right angle at C, $AC=9$ and $BC=12$. Each leg is lengthened by 3 and the angle at C stays a right angle. What is $\tan A$ in the new triangle?

A) $\frac{4}{5}$

B) $\frac{4}{3}$

C) $\frac{5}{4}$

D) $\frac{5}{3}$

Answer: (C) $\frac{5}{4}$

New legs: $AC=12$, $BC=15$. $\tan A=\frac{BC}{AC}=\frac{15}{12}=\frac{5}{4}$. Adding the same amount to each leg does not keep the triangle similar, so the ratio changes from $\frac{4}{3}$.

Trap: Students assume the trig ratio is unchanged ($\frac{4}{3}$).

---

22. GT-263 · Hard · Multiple choice · **No calculator needed**

A straight railway track up a hill has a constant grade of 75%, meaning it rises 75 meters for every 100 meters of horizontal distance. The track is 300 meters long, measured along the slope. What is the vertical rise of the track, in meters?

A) 400

B) 240

C) 180

D) 225

Answer: (C) 180

Grade is $\tan\theta = \frac{3}{4}$, so the sides are in ratio $3:4:5$ and $\sin\theta = \frac{3}{5}$. Rise $= 300 \cdot \frac{3}{5} = 180$.

Trap: Students treat the grade as the sine ($300 \cdot 0.75 = 225$) or give the horizontal run (240).

---

23. GT-264 · Hard · Multiple choice · **No calculator needed**

A drone flies 12 kilometers in a direction $60°$ north of east, then flies 12 kilometers due west. How far, in kilometers, is the drone from its starting point?

A) 12

B) 24

C) $12\sqrt{2}$

D) $6\sqrt{3}$

Answer: (A) 12

First leg ends at $(12\cos 60°, 12\sin 60°) = (6, 6\sqrt{3})$. After moving 12 west: $(-6, 6\sqrt{3})$. Distance $= \sqrt{36+108} = 12$.

Trap: Students assume the legs are perpendicular ($12\sqrt{2}$) or add them (24).

---

24. GT-265 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle B is $45°$ and the measure of angle C is $30°$. The altitude from A to $\overline{BC}$ has length 6. What is the length of $\overline{BC}$?

A) 12

B) $6+2\sqrt{3}$

C) $12+6\sqrt{2}$

D) $6+6\sqrt{3}$

Answer: (D) $6+6\sqrt{3}$

The altitude meets $\overline{BC}$ at D. In the $45°$ triangle, $BD = 6$. In the $30°$ triangle, $DC = \frac{6}{\tan 30°} = 6\sqrt{3}$. $BC = 6+6\sqrt{3}$.

Trap: Students multiply by $\tan 30°$ instead of dividing ($6+2\sqrt{3}$).

---

25. GT-266 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, the points $A(0,0)$, $B(6,3)$ and $C(4,7)$ are the vertices of a triangle. What is the value of $\tan(\angle CAB)$?

A) $\frac{1}{2}$

B) $\frac{2}{3}$

C) $\frac{3}{2}$

D) $\frac{7}{4}$

Answer: (B) $\frac{2}{3}$

Slopes: $AB = \frac{1}{2}$, $BC = \frac{7-3}{4-6} = -2$; their product is -1, so the right angle is at B. $AB = \sqrt{45}$, $BC = \sqrt{20}$, $\tan A = \frac{BC}{AB} = \sqrt{\frac{20}{45}} = \frac{2}{3}$.

Trap: Students give the slope of $\overline{AC}$ ($\frac{7}{4}$) or of $\overline{AB}$ ($\frac{1}{2}$) instead of a ratio of sides.

---

26. GT-267 · Medium · Multiple choice · **No calculator needed**

A parallelogram has sides of lengths 12 and 20. The cosine of one of its acute angles is 0.8. What is the area of the parallelogram?

A) 72

B) 144

C) 192

D) 240

Answer: (B) 144

$\sin\theta = \sqrt{1 - 0.8^2} = 0.6$. Height to the side of length 20 is $12 \cdot 0.6 = 7.2$. Area $= 20 \cdot 7.2 = 144$.

Trap: Students use the cosine as the height ratio (192) or ignore the angle (240).

---

27. GT-268 · Hard · Multiple choice · **No calculator needed**

From the top of a cliff 45 meters high, the angles of depression to two boats on opposite sides of the cliff are $30°$ and $45°$. The boats and the base of the cliff are on a straight line. What is the distance, in meters, between the boats?

A) $45\sqrt{3} - 45$

B) $45 + 15\sqrt{3}$

C) $45\sqrt{3}$

D) $45 + 45\sqrt{3}$

Answer: (D) $45 + 45\sqrt{3}$

Horizontal distances: $\dfrac{45}{\tan 30°} = 45\sqrt{3}$ and $\dfrac{45}{\tan 45°} = 45$. Opposite sides, so add: $45 + 45\sqrt{3}$.

Trap: Students subtract the distances (the same-side case) or multiply by $\tan 30°$.

---

28. GT-269 · Hard · Multiple choice · **No calculator needed**

In a right triangle, the measure of acute angle A is between $45°$ and $90°$. Which of the following is true?

A) $\cos A < \sin A < \tan A$

B) $\tan A < \sin A < \cos A$

C) $\sin A < \cos A < \tan A$

D) $\cos A < \tan A < \sin A$

Answer: (A) $\cos A < \sin A < \tan A$

Above $45°$ the opposite leg is longer than the adjacent leg, so $\cos A < \sin A$, and $\tan A = \dfrac{\sin A}{\cos A} > 1 > \sin A$. Check $60°$: $0.5 < 0.866 < 1.732$.

Trap: Students use the ordering that holds below $45°$ $(\sin A < \cos A)$.

---

29. GT-270 · Hard · SPR · **No calculator needed**

A rectangle has side lengths 6 and $6\sqrt{3}$. What is the measure, in degrees, of the acute angle formed by its diagonals?

Answer: 60

Each diagonal makes an angle with the longer side whose tangent is $\dfrac{6}{6\sqrt{3}} = \dfrac{1}{\sqrt{3}}$, so $30°$. The triangle formed by the longer side and the two half-diagonals has base angles $30°$ and apex angle $120°$. The adjacent angle at the centre, facing the shorter side, is $180° - 120° = 60°$.

Trap: Students give the diagonal-to-side angle $(30°)$ or the obtuse angle $(120°)$.

---

30. GT-271 · Hard · Multiple choice · **No calculator needed**

In the xy-plane, line ℓ has slope $\frac{3}{4}$ and line k is perpendicular to line ℓ. Line k makes an angle θ with the positive x-axis, measured counterclockwise, where $0° < \theta < 180°$. What is the value of $\cos\theta$?

A) $-\frac{4}{5}$

B) $\frac{3}{5}$

C) $-\frac{3}{5}$

D) $\frac{4}{5}$

Answer: (C) $-\frac{3}{5}$

Slope of k: $-\frac{4}{3}$, so θ is obtuse with direction $(-3, 4)$ (length 5). $\cos\theta = -\frac{3}{5}$.

Trap: Students drop the negative sign or use the sine ($\frac{4}{5}$).

---

31. GT-272 · Hard · Multiple choice · **No calculator needed**

Parallel lines ℓ and m are horizontal, with ℓ above m. Point P lies between the lines. Point A on ℓ and point B on m are both to the right of P. Segment $\overline{PA}$ makes a $38°$ angle with line ℓ, and segment $\overline{PB}$ makes a $47°$ angle with line m, both angles opening toward P. What is the measure, in degrees, of angle APB?

A) 42.5

B) 95

C) 85

D) 9

Answer: (C) 85

Draw a line through P parallel to ℓ and m. Alternate interior angles give $38°$ above it and $47°$ below it, so $\angle APB = 38° + 47° = 85°$.

Trap: Students compute $180 - (38 + 47) = 95$ or the difference 9.

---

32. GT-273 · Hard · Multiple choice · **No calculator needed**

In triangle ABC, the measure of angle A is $52°$. The bisectors of angles B and C intersect at point I. What is the measure, in degrees, of angle BIC?

A) 116

B) 128

C) 64

D) 104

Answer: (A) 116

$B + C = 128°$, so $\frac{B}{2} + \frac{C}{2} = 64°$ and $\angle BIC = 180° - 64° = 116°$.

Trap: Students compute $180 - 52 = 128$, or $2 \cdot 52 = 104$, or stop at 64.

---

33. GT-274 · Medium · Multiple choice · **No calculator needed**

In triangle ABC, $AB = 2x + 1$, $BC = x + 6$ and $CA = 3x - 4$. The perimeter of the triangle is 45. Which angle of the triangle has the greatest measure?

A) Angle A

B) It cannot be determined from the information given.

C) Angle C

D) Angle B

Answer: (D) Angle B

$6x + 3 = 45 \Rightarrow x = 7$: $AB = 15$, $BC = 13$, $CA = 17$. The greatest angle is opposite the longest side $\overline{CA}$, which is angle B.

Trap: Students name a vertex of the longest side (A or C) instead of the opposite angle.

Problem-Solving and Data Analysis (35 items)

34. PS-176 · Hard · Multiple choice · **No calculator needed**

If 18% of a is 24, what is $a\%$ of 18?

A) 4.32

B) 24

C) $2{,}400$

D) $133\frac{1}{3}$

Answer: (B) 24

$a\%$ of $18 = \dfrac{a \cdot 18}{100} = 18\%$ of $a = 24$. No need to find a.

Trap: Students solve for a ($133\frac{1}{3}$) and stop, or multiply $0.18 \cdot 24$.

---

35. PS-177 · Medium · Multiple choice · **Calculator**

The monthly rent of an apartment increased by 60%. By what percent must the new rent decrease to return to the original rent?

A) 62.5%

B) 37.5%

C) 60%

D) 40%

Answer: (B) 37.5%

New rent $= 1.6R$. Decrease needed: $\dfrac{0.6R}{1.6R} = 0.375 = 37.5\%$.

Trap: Students reuse 60% or compute $100 - 60 = 40$.

---

36. PS-178 · Hard · Multiple choice · **Calculator**

A store's profit on each jacket is 37.5% of the selling price. The profit is what percent of the cost of the jacket to the store?

A) 37.5%

B) 27.3%

C) 62.5%

D) 60%

Answer: (D) 60%

Let the price be 100: profit 37.5, cost 62.5. $\dfrac{37.5}{62.5} = 0.6 = 60\%$.

Trap: Students keep the selling price as the base (37.5%) or compute cost as a percent of price (62.5%).

---

37. PS-179 · Hard · Multiple choice · **Calculator**

A restaurant adds a 15% tax and a 10% service charge to a bill. Both are calculated on the original subtotal. The total paid is 250 dollars. What was the subtotal, in dollars?

A) 200.00

B) 217.39

C) 197.63

D) 187.50

Answer: (A) 200.00

Both percents use the same base, so they add: total $= 1.25 \cdot$subtotal. Subtotal $= \dfrac{250}{1.25} = 200$.

Trap: Students chain the percents ($\frac{250}{1.15 \cdot 1.10} \approx 197.63$) or ignore one charge.

---

38. PS-180 · Hard · Multiple choice · **No calculator needed**

A club has 40 members, and 30% of the members are girls. How many girls must join the club, with no one leaving, so that girls make up 60% of the members?

A) 20

B) 12

C) 30

D) 28

Answer: (C) 30

Girls now: 12. $12 + g = 0.6(40 + g) \Rightarrow 0.4g = 12 \Rightarrow g = 30$.

Trap: Students take 60% of the old total ($24 - 12 = 12$) and forget the total grows.

---

39. PS-181 · Hard · Multiple choice · **Calculator**

A worker spends 30% of her monthly salary on rent, then 25% of the remaining amount on food, then 40% of what is left after that on savings. She has $3{,}780$ dollars left. What is her monthly salary, in dollars?

A) $75{,}600$

B) $7{,}200$

C) $12{,}000$

D) $6{,}300$

Answer: (C) $12{,}000$

Remaining fraction: $0.7 \cdot 0.75 \cdot 0.6 = 0.315$. Salary $= \dfrac{3{,}780}{0.315} = 12{,}000$.

Trap: Students add the percents (95% spent) or skip a step.

---

40. PS-182 · Hard · Multiple choice · **Calculator**

Two positive numbers a and b are in the ratio $a : b = 9 : 5$. The number b is what percent less than a, to the nearest tenth of a percent?

A) 44.4%

B) 80.0%

C) 55.6%

D) 28.6%

Answer: (A) 44.4%

$\dfrac{a - b}{a} = \dfrac{4}{9} \approx 44.4\%$.

Trap: Students use b as the base (80%, which is how much greater a is) or use the total parts.

---

41. PS-183 · Hard · Multiple choice · **No calculator needed**

The number y is $p\%$ less than the positive number x, where $0 < p < 100$. Which expression is equal to x?

A) $\dfrac{(100 - p)y}{100}$

B) $\dfrac{100y}{100 + p}$

C) $y\left(1 + \dfrac{p}{100}\right)$

D) $\dfrac{100y}{100-p}$

Answer: (D) $\dfrac{100y}{100-p}$

$y = x\left(1-\dfrac{p}{100}\right) = x\cdot\dfrac{100-p}{100}$, so $x = \dfrac{100y}{100-p}$. Check $p=20$, $y=80$: $x=100$.

Trap: Students add $p\%$ back to y (using the wrong base).

---

42. PS-184 · Hard · Multiple choice · **No calculator needed**

Fresh dates are 80% water by mass. After drying, the dates are 50% water by mass. How many kilograms of dried dates are produced from 60 kilograms of fresh dates?

A) 30

B) 24

C) 42

D) 12

Answer: (B) 24

The non-water part does not change: $0.2\cdot 60 = 12$ kg. After drying it is 50% of the mass: $\dfrac{12}{0.5} = 24$ kg.

Trap: Students subtract 30 percentage points of the mass ($60\cdot 0.7 = 42$) or halve the mass.

---

43. PS-185 · Hard · Multiple choice · **No calculator needed**

In a survey, exactly 37.5% of the people answered "yes," exactly 20% answered "no," and the rest were undecided. What is the least possible number of people surveyed?

A) 5

B) 40

C) 8

D) $1{,}000$

Answer: (B) 40

$37.5\% = \dfrac{3}{8}$ and $20\% = \dfrac{1}{5}$, so the number surveyed must be a multiple of 8 and of 5: least is 40 (15 yes, 8 no, 17 undecided).

Trap: Students satisfy only one percent (8 or 5).

---

44. PS-186 · Hard · Multiple choice · **No calculator needed**

At a school, 85% of the boys and 60% of the girls passed an exam, and 70% of all the students passed. What percent of the students at the school are boys?

A) 25%

B) 50%

C) 60%

D) 40%

Answer: (D) 40%

Let b be the fraction of boys: $0.85b + 0.6(1-b) = 0.7 \Rightarrow 0.25b = 0.1 \Rightarrow b = 0.4 = 40\%$.

Trap: Students assume equal groups (50%) or swap the groups (60%).

---

45. **PS-187** · Medium · Multiple choice · **Calculator**

A runner's pace is 6 minutes 15 seconds per kilometer. What is the runner's speed, in kilometers per hour?

A) 9.6

B) 6.25

C) 9.76

D) 0.16

Answer: (A) 9.6

6 min 15 s $= 6.25$ min. Speed $= \dfrac{60}{6.25} = 9.6$ km/h.

Trap: Students read 6 min 15 s as 6.15 min (≈ 9.76).

---

46. **PS-188** · Hard · Multiple choice · **No calculator needed**

A pipe can fill an empty tank in 6 hours, and a drain can empty the full tank in 9 hours. Starting with the tank empty, the pipe and the drain are both open for 4 hours, and then the drain is closed. How many more hours does it take the pipe to fill the tank?

A) 2

B) $\dfrac{26}{3}$

C) $\dfrac{14}{3}$

D) 14

Answer: (C) $\dfrac{14}{3}$

Net rate $\dfrac{1}{6} - \dfrac{1}{9} = \dfrac{1}{18}$ tank per hour; after 4 hours: $\dfrac{2}{9}$ full. Remaining $\dfrac{7}{9}$ at $\dfrac{1}{6}$ per hour: $\dfrac{7/9}{1/6} = \dfrac{14}{3}$ hours (4 h 40 min).

Trap: Students give the total time ($8\frac{2}{3}$ h) or leave the drain open.

---

47. **PS-189** · Hard · Multiple choice · **No calculator needed**

A crew of 12 workers can finish a job in 20 days. After the crew has worked for 5 days, 3 more workers join, all working at the same rate. How many days in total does the job take?

A) 16

B) 25

C) 17

D) 12

Answer: (C) 17

Total work $= 12 \cdot 20 = 240$ worker-days. Done in 5 days: 60. Remaining 180 with 15 workers: 12 days. Total $5 + 12 = 17$.

Trap: Students give only the remaining days (12) or assume 15 workers from the start (16).

---

48. **PS-190** · Hard · Multiple choice · **No calculator needed**

A driver travels from one city to another at an average speed of 60 kilometers per hour. At what average speed, in kilometers per hour, must the driver return along the same road so that the average speed for the whole round trip is 80 kilometers per hour?

A) 120

B) 106.7

C) 160

D) 100

Answer: (A) 120

Let the distance be d. $\dfrac{2d}{d/60 + d/v} = 80 \Rightarrow \dfrac{2}{80} = \dfrac{1}{60} + \dfrac{1}{v} \Rightarrow \dfrac{1}{v} = \dfrac{1}{120}$, $v = 120$.

Trap: Students average the speeds (100).

---

49. **PS-191** · Hard · Multiple choice · **No calculator needed**

A printer prints p pages every m seconds. Which expression gives the number of minutes the printer takes to print n pages?

A) $\dfrac{60m}{pn}$

B) $\dfrac{60mn}{p}$

C) $\dfrac{pn}{60m}$

D) $\dfrac{mn}{60p}$

Answer: (D) $\dfrac{mn}{60p}$

Seconds per page $= \dfrac{m}{p}$; n pages take $\dfrac{mn}{p}$ seconds $= \dfrac{mn}{60p}$ minutes. Check $p = 10$, $m = 5$, $n = 120$: 60 s $= 1$ min.

Trap: Students multiply by 60 instead of dividing, or invert the rate.

---

50. **PS-192** · Hard · Multiple choice · **Calculator**

A bank sells US dollars at 3.77 riyals per dollar. It first deducts a 1% commission from the riyals a customer pays. How many US dollars does a customer receive for $5{,}655$ riyals?

A) $1{,}515$

B) $1{,}485$

C) $21{,}319$

D) $1{,}500$

Answer: (B) $1{,}485$

Riyals exchanged: $0.99 \cdot 5{,}655 = 5{,}598.45$. Dollars $= \dfrac{5{,}598.45}{3.77} = 1{,}485$.

Trap: Students ignore the commission $(1{,}500)$, add it, or multiply by the rate.

---

51. **PS-193** · Hard · Multiple choice · **No calculator needed**

The data set $3, 5, 6, 8, 9, 11, 12, 46$ has one outlier, 46, which is removed. Which statement about the effect of removing 46 is true?

A) Only the mean changes.

B) Both the mean and the median decrease, and the mean decreases more.

C) The mean decreases and the median increases.

D) The mean decreases and the median stays the same.

Answer: (B) Both the mean and the median decrease, and the mean decreases more.

Mean: $12.5 \to \dfrac{54}{7} \approx 7.71$. Median: $\dfrac{8+9}{2} = 8.5 \to 8$ (the count changes from 8 to 7). Both decrease, and the mean decreases more.

Trap: Students think the median cannot change when an outlier is removed.

---

52. PS-194 · Hard · Multiple choice · **No calculator needed**

A data set consists of $12, 15, 18, 21, 24, 27, 30, 33, 36, 40$. The value 40 is mistakenly recorded as 400. How do the median, the interquartile range (IQR) and the standard deviation of the recorded data compare with those of the correct data?

A) Only the median is unchanged.

B) The median, IQR and standard deviation are all unchanged.

C) The median is unchanged, but the IQR and standard deviation are greater.

D) The median and IQR are unchanged, but the standard deviation is greater.

Answer: (D) The median and IQR are unchanged, but the standard deviation is greater.

Only the maximum changed. The median (25.5) and the quartiles $(18$ and $33)$ do not involve the maximum, so the median and IQR are unchanged. The standard deviation measures distance from the mean and grows greatly (about 8.8 to about 113).

Trap: Students think every statistic changes with an outlier, or that the IQR is affected.

---

53. PS-195 · Hard · Multiple choice · **No calculator needed**

Section P has 60 students and section Q has 24 students. Test-score summaries: Section P: minimum 55, first quartile 68, median 76, third quartile 84, maximum 97. Section Q: minimum 62, first quartile 77, median 83, third quartile 88, maximum 95. Which statement must be true?

A) At least 75% of the students in Q scored above the median of P.

B) The IQR of Q is greater than the IQR of P.

C) More students in Q than in P scored above 84.

D) The range of Q is greater than the range of P.

Answer: (A) At least 75% of the students in Q scored above the median of P.

In Q, at least 75% of the scores are at or above its first quartile, 77, which is above P's median, 76. the statement about more students above 84 cannot be decided (different section sizes; only percents are known); the range of Q (33) is less than that of P (42); the IQR of Q (11) is less than that of P (16).

Trap: Students compare counts when only percents are known, or compare ranges by eye.

---

54. PS-196 · Hard · Multiple choice · **No calculator needed**

At a company, the median monthly salary is $9{,}000$ dollars and the mean monthly salary is $14{,}500$ dollars. Which statement must be true?

A) The most common salary is $9{,}000$ dollars.

B) The standard deviation of the salaries is $5{,}500$ dollars.

C) No more than half of the employees earn more than $14{,}500$ dollars.

D) More than half of the employees earn more than $14{,}500$ dollars.

Answer: (C) No more than half of the employees earn more than $14{,}500$ dollars.

Any employee earning more than $14{,}500$ also earns more than the median $9{,}000$, and at most half of the employees earn more than the median. So no more than half earn more than $14{,}500$.

Trap: Students read the mean as the typical salary, or treat mean minus median as the standard deviation.

---

55. PS-197 · Hard · SPR · **No calculator needed**

In a data set, the values 1, 2, 3, 4 and 5 occur with relative frequencies 10%, 15%, 20%, 30% and 25%, respectively. What is the median of the data set minus the mean of the data set?

Answer: 0.55 **(or** $\frac{11}{20}$**)**

Cumulative: $10\%, 25\%, 45\%, 75\%$, so the 50% point is at the value 4: median $= 4$. Mean $= 0.1{+}0.3{+}0.6{+}1.2{+}1.25 = 3.45$. Difference $= 0.55$.

Trap: Students think the total count is needed, or take the median of the percents.

---

56. PS-198 · Hard · Multiple choice · **No calculator needed**

A data set has six values. Their deviations from 40 (value minus 40) are -7, 3, 12, -5, x and 2. The mean of the data set is 42. What is the value whose deviation from 40 is x?

A) 35

B) 7

C) 47

D) 52

Answer: (C) 47

The deviations from 40 sum to $6(42-40)=12$, not 0 (they sum to 0 only around the mean). $5+x=12$, $x=7$; the value is 47.

Trap: Students set the deviations' sum to 0 ($x=-5$, value 35) or stop at $x=7$.

---

57. PS-199 · Hard · Multiple choice · **No calculator needed**

A data set of 30 values has a mean of 50 and a standard deviation of 10. The values 38 and 62 are added to the data set. Which statement is true?

A) The mean is unchanged and the standard deviation increases.

B) The mean is unchanged and the standard deviation decreases.

C) The mean and the standard deviation are both unchanged.

D) The mean and the standard deviation both increase.

Answer: (A) The mean is unchanged and the standard deviation increases.

The pair is symmetric about 50, so the mean stays 50. Each new value is 12 from the mean, farther than the typical distance 10, so the standard deviation increases (to about 10.1).

Trap: Students assume values inside the range lower the SD, or that a symmetric pair leaves it unchanged.

---

58. PS-200 · Medium · SPR · **No calculator needed**

In a data set, a value is called a low outlier if it is less than $Q_1 - 1.5 \cdot \text{IQR}$, where Q_1 is the first quartile and IQR is the interquartile range. For a data set, $Q_1 = 28$ and the third quartile is 40. What is the greatest integer value that would be a low outlier?

Answer: 9

IQR $= 12$; fence $= 28 - 18 = 10$. A low outlier is strictly less than 10, so the greatest integer is 9.

Trap: Students include the boundary (10) or forget the factor 1.5 (16).

---

59. PS-201 · Hard · Multiple choice · **No calculator needed**

The recorded values of a data set are $18, 22, 27, 30, 33, 36, 41, 47, 52$. The value 52 should have been 25. After the correction, how do the mean and the median change?

A) The mean decreases by 3 and the median is unchanged.

B) The median decreases and the mean is unchanged.

C) The mean decreases by 27 and the median decreases by 3.

D) The mean and the median both decrease by 3.

Answer: (D) The mean and the median both decrease by 3.

Mean: $34 \to 31$ (the sum drops by 27 over 9 values). Median: $33 \to 30$, because 25 moves below the middle and the middle value shifts. Both decrease by 3.

Trap: Students assume the median is unchanged when one value is corrected.

---

60. **PS-202** · Hard · Multiple choice · **No calculator needed**

A company randomly selected 200 of its employees and gave them new office chairs. Their average productivity rose 8% during the next 3 months. Which conclusion is best supported?

A) The new chairs caused the rise for all employees.

B) The rise cannot be attributed to the chairs, because there was no comparison group.

C) The new chairs had no effect on productivity.

D) The new chairs caused the rise for the 200 employees only.

Answer: (B) The rise cannot be attributed to the chairs, because there was no comparison group.

There was no comparison group without new chairs, so the rise could come from other changes over the same months. Random selection allows generalization, not a causal claim.

Trap: Students think a random sample alone supports cause and effect.

---

61. **PS-203** · Hard · Multiple choice · **No calculator needed**

In an experiment, garden plots were randomly assigned to receive 0, 20 or 40 grams of a fertilizer. The plots given 40 grams had the highest mean yield. Which conclusion is best supported?

A) Doubling the fertilizer doubles the yield.

B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.

C) No causal conclusion is possible.

D) Using 80 grams would give an even higher yield.

Answer: (B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.

Random assignment supports a causal conclusion, but only within the tested doses and for plots like these.

Trap: Students extrapolate a valid causal result beyond the tested doses.

---

62. **PS-204** · Hard · Multiple choice · **No calculator needed**

A survey was mailed to $1{,}200$ randomly selected households, and 180 surveys were returned. Of these, 72% supported a new fee. What is the main reason the 72% may not represent all households?

A) There was no random assignment to treatments.

B) The households were selected at random.

C) The sample of $1{,}200$ is too small.

D) Many selected households did not respond, and those who responded may differ from those who did not.

Answer: (D) Many selected households did not respond, and those who responded may differ from those who did not.

Only 15% responded. People who choose to return a survey may differ from those who do not (non-response bias), even though the original selection was random.

Trap: Students think random selection guarantees a representative result.

---

63. **PS-205** · Medium · Multiple choice · **No calculator needed**

An observational study found that children from homes with more books tend to have higher reading scores. Which of the following is the most plausible confounding variable?

A) Parents' education level

B) The scale used for the reading test

C) The number of pages per book

D) The number of children in the study

Answer: (A) Parents' education level

A confounder is related to both variables: parents' education level can affect both the number of books at home and the children's reading scores.

Trap: Students choose a variable linked to only one of the two variables, or a feature of the study design.

---

64. PS-206 · Medium · Multiple choice · **No calculator needed**

To estimate the average weekly exercise time of adults in a city, a researcher randomly selected 400 people from the membership lists of the city's gyms. Why is the estimate likely to be biased?

A) The sample size is too small.

B) The selection was random, so the estimate is not biased.

C) The list leaves out adults who are not gym members.

D) There was no random assignment.

Answer: (C) The list leaves out adults who are not gym members.

The sampling frame (gym members) leaves out adults who are not gym members, who probably exercise less. Random selection from a biased list is still biased.

Trap: Students accept random selection without checking the list it came from.

---

65. PS-207 · Medium · Multiple choice · **No calculator needed**

A teacher randomly selected 50 students and asked each one, face to face, "Have you ever copied homework?" Of the students, 6% answered yes. Which is the most likely problem with this result?

A) The sample was not selected at random.

B) There was no control group.

C) Students may give untrue answers because the question is sensitive and asked in person.

D) A sample of 50 students is too large.

Answer: (C) Students may give untrue answers because the question is sensitive and asked in person.

The question is sensitive and asked in person by a teacher, so students may under-report (response bias), even though the sample was random.

Trap: Students check only how the sample was chosen and miss the measurement problem.

---

66. PS-208 · Medium · Multiple choice · **No calculator needed**

For a data set, the correlation coefficient between height in centimeters and mass in kilograms is $r = 0.72$. The heights are converted to inches, the masses to pounds, and the two variables are swapped between the axes. What is the correlation coefficient for the converted data?

A) 0.72

B) -0.72

C) $0.72 \cdot \dfrac{2.2}{2.54}$

D) $\dfrac{1}{0.72}$

Answer: (A) 0.72

Correlation has no units and is symmetric: changing units (positive scale factors) and swapping the axes leave r unchanged, 0.72.

Trap: Students scale r by the conversion factors or take its reciprocal.

---

67. PS-209 · Hard · Multiple choice · **No calculator needed**

For data set P, the line of best fit is $\hat{y} = 4.2x + 3$ and the correlation coefficient is 0.41. For data set Q, the line of best fit is $\hat{y} = 0.6x + 10$ and the correlation coefficient is 0.93. Which statement is true?

A) The associations cannot be compared because the slopes are different.

B) P has the stronger linear association because its slope is greater.

C) Q has a weaker association because its slope is less than 1.

D) Q has the stronger linear association.

Answer: (D) Q **has the stronger linear association.**

The strength of a linear association is measured by r, not by the slope. Q has r closer to 1, so its points fit a line more closely, relative to the spread of the data.

Trap: Students think a steeper slope means a stronger association.

---

68. **PS-210** · Hard · Multiple choice · **No calculator needed**

A model for a data set is $\hat{y} = 2.5x + 10$. The data points are $(2, 17)$, $(4, 19.5)$, $(6, 24)$, $(8, 31)$ and $(10, 34.5)$. For how many of the data points does the model underestimate the actual value of y?

A) 5

B) 2

C) 3

D) 1

Answer: (B) 2

Residuals (actual $-$ predicted): $17 - 15 = 2$, $19.5 - 20 = -0.5$, $24 - 25 = -1$, $31 - 30 = 1$, $34.5 - 35 = -0.5$. The model underestimates when the residual is positive: 2 points.

Trap: Students reverse "underestimate" and count the negative residuals (3).
